

Transformation without loss

Residual cohomology of token-configuration arenas under change of substrate, symmetry, and admissibility

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Lead-in

Put some indistinguishable tokens on the vertices of a finite graph, one per vertex, and let a token step to an empty neighbour. That is the whole system. The graph is the *substrate*; the space of configurations and legal steps is the *arena*.

Now change something about it — add an edge, subdivide one, glue two vertices together, allow a site to hold two tokens instead of one, or simply look at the system with a coarser symmetry. Each of those changes alters which transformations exist. The question this paper asks is narrow and answerable:

When the rules of movement change, what happens to the structure that was there before?

The naive answers are that structure is created or destroyed. Both turn out to be wrong in a precise and checkable sense. Every elementary change either *extends* the transformation structure, *re-expresses* it, or *hides* it from a particular observer. Structure is never annihilated. The one operation that does produce a genuine kernel — restricting which configurations are admissible — produces it as a **relative cohomology class**: something present in the larger admissible space and absent from the smaller one, with an exact dimension formula.

The headline object is a single integer attached to each such restriction. For a distinguished site allowed to hold r tokens instead of $r + 1$, that integer is the first Betti number of a small quotient graph, and it can be computed without ever building the arena.

A concrete instance, which the reader can verify in Section 9: on the 7-cycle with four tokens, the number of independent transformation channels visible to a symmetric observer goes $16 \rightarrow 27 \rightarrow 31 \rightarrow 31$ as the distinguished site's capacity is relaxed from one token to four. The increments 11,4,0 are the capacity defects. The final zero is **not** a saturation effect — it is a statement about the local symmetry at that site, and on other substrates it is nonzero.

o. Scope

This is a mathematics paper. It is part of a larger programme (NEXUS) whose ambition is explanatory rather than descriptive: not another representation of physical regularity, but an account of what transformation requires. That

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

ambition does not appear in the theorems below and is not used to support any of them. The rule followed throughout is **prove downward, interpret upward, never prove upward by interpretation**. Section 11 is the only interpretive section and proves nothing.

One point of orientation, because it affects how the results should be read. A complete explanation of transformation is not the same thing as a terminal state of the system being explained. The objects here are finite and the theorems are closed; the programme they belong to expects its own descriptions to keep transforming. Nothing in Sections 1–10 depends on that expectation.

1. Setup

$X = (V, E)$ is a finite connected simple graph, $K \geq 1$ a token count, $G \leq \text{Aut}(X)$.

Arena. $\mathcal{A}(X, K)$ has as vertices the K -subsets $S \subseteq V$ and as edges the moves $S \rightarrow S - u + w$ with $uw \in E$, $u \in S$, $w \notin S$.

Lemma 1.1 (edge parametrisation). Arena edges are in G -equivariant bijection with pairs (e, T) , $e = \{u, w\} \in E$, $T \subseteq V \setminus e$, $|T| = K - 1$. The two orientations are the two choices of which endpoint of e carries the token at the source.

Proof. $T = S \cap S'$ and e is the moved token's base edge; the correspondence is manifestly equivariant. ▫

Everything downstream follows from this. It is the reason the base graph, not the arena, controls most of the answers.

Residual. With C^0, C^1 the real cochain groups of the arena 1-skeleton (C^1 antisymmetric under edge reversal),

$$R_G(X, K) = \frac{(C^1)^G}{d_0((C^0)^G)}, \quad \epsilon(X, K; G) = \dim R_G(X, K).$$

Averaging over the finite group gives $R_G \cong H^1(\mathcal{A})^G$ over $\mathbb{R}$; the quotient presentation is the one used below. Three distinct objects appear in the literature on this and must not be conflated: $H^1(\mathcal{A}; \mathbb{R})$, its invariant part R_G , and H^1 of the complex with commuting squares filled. Every statement here names which one it means.

2. What ϵ is

Theorem 2.1. $\dim(C^1)^G = e_G^{\text{ns}}$, the number of G -orbits of arena edges whose stabiliser contains no orientation-reversing element.

Proof. Orbit–stabiliser on oriented edges. If some g reverses e then invariance forces $\omega(e) = -\omega(e)$, hence $\omega(e) = 0$; otherwise one scalar determines ω on the whole orbit. ▫

Theorem 2.2. For connected $\mathcal{A}$, $\dim d_0((C^0)^G) = v_G - 1$, v_G the number of configuration orbits. Hence

$$\boxed{\epsilon = e_G^{\text{ns}} - v_G + 1 = \beta_1(\mathcal{A}/G)}$$

where $\mathcal{A}/G$ is the quotient graph with the reversible edge orbits deleted.

The three descriptions — orbit counting, cochain dimensions, quotient-graph topology — are the same integer. Measured across 30 arena/predicate combinations with no discrepancy.

Theorem 2.3 (closed form). Writing $P_g(x) = \prod_{c \in \text{Cyc}(g)} (1 + x^{|c|})$, $m^+(g)$ for edges of X fixed pointwise by g and $m^-(g)$ for edges whose endpoints g swaps,

$$\epsilon(X, K; G) = 1 + \frac{1}{|G|} \sum_{g \in G} (m^+(g) [x^{K-1}] P_g / (1+x)^2 - m^-(g) [x^{K-1}] P_g / (1+x^2 - [x^K] P_g))$$

Cycle structure of G on $V(X)$ plus two edge counts per group element. No arena is built. Verified against direct orbit counting on 106 (substrate, K , block-profile) cases across five base graphs including a non-vertex-transitive one, $K = 2..5$: no discrepancies.

Two immediate consequences. $\epsilon(K_N, K) = 0$ for all N, K under the full symmetric group — S_N is transitive on (e, T) and the transposition $(u\ w)$ reverses e while fixing T pointwise, so the single edge orbit is reversible. And for the trivial group there is no quotient at all, so $\epsilon(X, K; \{1\}) = \beta_1(\mathcal{A}(X, K))$ exactly — not a limit.

Measured, not proved: $\mathcal{H}_G^1 = 0$. Across three arenas and both standard 2-cell predicates (disjoint-support commuting squares; all arena 4-cycles), the invariant harmonic space was zero in every case, so the entire invariant non-exact sector is coexact. Cell counts moved by a factor of 5.7 without moving ϵ by one dimension: the residual is a property of the 1-skeleton and the group, and the 2-cells only decide how it splits. No case with $\mathcal{H}_G^1 \neq 0$ is known.

3. What a reader can see

A *reader* coarse-grains the substrate into blocks and records block-to-block flow. Two natural responses exist, and both are blind to the residual, for different reasons.

The scalar response $Q_\ell = d\psi_\ell, \psi_\ell(S) = \sum_B |B \cap S|^2$, is exact, so its integral round any loop vanishes — in the arena and in any quotient of it.

The matrix response is subtler. Contracting the antisymmetric block-flow table with any fixed antisymmetric M gives $2M_{b(u)b(w)}$ for a move $u \rightarrow w$: it depends only on the moved **base edge and its orientation**. Hence every matrix-reader cochain is a pullback $\pi^*\eta$ along the equivariant map π from arena directed edges to base directed edges.

Theorem 3.1 (reader bound). $\dim \text{proj}_R(\pi^* C^1(X)) \leq \epsilon(X, 1; G)$.

Proof. Equivariance gives $\langle r, \pi^*\eta \rangle = \langle r, \pi^*(\Pi_X \eta) \rangle$, so only invariant base forms contribute. And $\pi^*(d_X f) = d_{\mathcal{A}} F$ with $F(S) = \sum_{v \in S} f(v)$, so exact base forms pull back to coboundaries. What remains is the base graph's own residual at one token, and $\mathcal{A}(X, 1) = X$. ◻

Measured: the bound is attained and is independent of K . Nineteen cases; on every symmetric substrate the reach is 0 — 0 of 2 channels on C_8 at $K = 3$, 0 of 3 at $K = 4$, 0 of 1 on Q_3 and Petersen — and on deliberately constructed trivial-automorphism bases with $\epsilon(X, 1) = 1, 2$ the reach is exactly 1,1,1 and 2,2,2 across $K = 2, 3, 4$ against arena residuals of 10,17,10 and 14,23,14.

$\epsilon(X, K) - \epsilon(X, 1) =$ channels created by K tokens sharing a substrate, unreachable by any one-token observable

4. The four motions

Let $X \rightarrow X'$ be an elementary change. The residual is only defined relative to a group, and the group changes too, so comparisons pass through $G_\cap = \text{Aut}(X) \cap \text{Aut}(X')$. There is no map $R_X \rightarrow R_{X'}$; there is a span.

motion	what happens to a transformation	induced map on residuals	measured
extension — add a base edge	survives verbatim; new ones appear	Δ_R onto, coker = 0	6 pairs
re-factorisation — subdivide an edge	$[u_0 \rightarrow w_0] \mapsto [u_0 \rightarrow z] + [z \rightarrow w_0]$	F_R^* onto, coker = 0	5 pairs, $\partial F = F \partial$ to 0.0
hiding — enlarge the group	unchanged	$R_{G'} \hookrightarrow R_G$, ker = 0	5 pairs
identification — contract an edge	configurations merge, q many-to-one	q^* injective; kernel only via admissibility	Section 5

No motion removes a channel. The one case that looks like removal — $P_6 \rightarrow C_6$ takes the residual from 5 to 0 — added six transformations and destroyed none: the five channels persist as $\mathbb{Z}_2$ -invariant cochains, and $R_{D_6} = 0$ is a statement about the enlarged observer.

The same correction applies to a number that motivated much of this work. $\epsilon(C_N, 3) = 0, 1, 2, 3$ for $N = 6, 7, 8, 9$ looks like loops appearing as the cycle lengthens. Pulling the C_7 channel back along the subdivision chain map gives rank 1 of 1, landing inside R_{C_6} at the common group; likewise 2 of 2 and 3 of 3. The cochains are present throughout. The shorter cycle's dihedral group cannot see them.

edits extend or re-factor; symmetry hides; projection erases visibility, not structure

5. Admissibility: the capacity filtration

Fix a distinguished site $m \in V$ and let $G = \text{Stab}_{\text{Aut}(X)}(m)$. Let $X^{(r)}$ be the arena in which m may hold up to r tokens and every other site at most one. Then $X^{(r)} \subseteq X^{(r+1)}$, with restriction $\sigma_r: R_G(X^{(r+1)}) \rightarrow R_G(X^{(r)})$ and newly admitted layer

$$\mathcal{D}_r = \{m^{r+1} + U : U \subseteq V \setminus \{m\}, |U| = K - r - 1\}.$$

Theorem 5.1. σ_r is surjective; $\text{coker} \sigma_r = 0$.

Proof. Extend an invariant 1-cochain by 0 on the new edges and average; the extension is invariant and restricts correctly. For a graph $C^2 = 0$, so every 1-cochain is a cocycle and the extension descends. ▀

Definition. $D_r(G, X, K) = \ker \sigma_r$, the **capacity defect**, and $p_r = \dim D_r$.

Theorem 5.2. $D_r \cong H_G^1(X^{(r+1)}, X^{(r)})$.

Proof. If $[\omega] \in \ker \sigma_r$, then $\omega|_{X^{(r)}} = d_0 f$ with f invariant; extend f and average to $\tilde{f}$, which restricts to f because G preserves the layers. Then $\omega - d_0 \tilde{f}$ vanishes on $X^{(r)}$, so every kernel class has a representative supported on the new layer. The converse is immediate. ▫

5.1 The cone theorem

The relative complex is the reduced complex of the quotient space: $X^{(r)}$ is **collapsed to a single point $*$** , not deleted. Define the graph Q_r :

- vertices: the G -orbits of $\mathcal{D}_r$, **plus the cone point $*$** ;
- edges: the G -orbits of arena moves with at least one endpoint in $\mathcal{D}_r$ (cross moves land on $*$; internal moves join two layer orbits), with the reversible orbits deleted.

Theorem 5.3 (cone). $\dim D_r = \beta_1(Q_r) = |E_Q| - |V_Q| + c_Q$.

Proof. By Theorem 2.1 applied relatively, $\dim C_G^1 = |E_Q|$. Relative 0-cochains are functions on the non-cone vertices, so $\dim C_G^0 = |V_Q| - 1$. The relative coboundary is the ordinary graph coboundary with the value at $*$ pinned to zero, so its kernel is the constants on each component with the $*$ -component forced to zero: $\dim \ker = c_Q - 1$, hence $\text{rank} = |V_Q| - c_Q$. Since $C^2 = 0$, $\dim D_r = |E_Q| - (|V_Q| - c_Q)$. ▫

Since $|V_Q| = v_{G, \mathcal{D}_r} + 1$ and $|E_Q| = e_{G, \mathcal{D}_r}^{ns}$:

$$\dim D_r = e_{G, \mathcal{D}_r}^{ns} - v_{G, \mathcal{D}_r} - 1 + c_Q, \quad \text{and} \quad D_r = 0 \Leftrightarrow Q_r \text{ is a forest.}$$

The cone point is not bookkeeping. Without it the saturated case would return $\beta_1 = -1$.

5.2 Connectivity is automatic

Lemma 5.4 (reversible orbits are loops). If $g \in G$ reverses the arena edge $\{S, T\}$ then $gS = T$, so $[S]_G = [T]_G$ and the orbit is a self-loop in the quotient. Deleting self-loops cannot change components, so $c(Q_r) = c(\hat{Q}_r)$, the count before deletion. ▫

Lemma 5.5 (vacancy propagation). Let $S = m^{r+1} + U$ with a vacancy $v \notin \{m\} \cup U$. If $N(m) \subseteq U$ there is no immediate move to $X^{(r)}$, but along a substrate path from v to $N(m)$ the first occupied-to-vacant edge $x \rightarrow y$ has $x \neq m$, so the move preserves the occupancy $r + 1$ and stays inside $\mathcal{D}_r$. Iterating frees a neighbour of m ; then $m \rightarrow v_j$ lands in $X^{(r)}$. ▫

$$\boxed{N(m) \subseteq U \Rightarrow c(Q_r) > 1}$$

Proposition 5.6 (saturation is degenerate). $c(Q_r) > 1$ requires $U = V \setminus \{m\}$, i.e. $K = |V| + r$. But a configuration of $X^{(r)}$ needs $|U| = K - c \leq |V| - 1$ with $c \leq r$, forcing $c \geq K - |V| + 1 = r + 1$, a contradiction. Hence $K = |V| + r$ implies $X^{(r)} = \emptyset$. ▫

So whenever the old arena is nonempty, $c(Q_r) = 1$ and

$$\boxed{\dim D_r = e_{G, \mathcal{D}_r}^{ns} - v_{G, \mathcal{D}_r}}$$

An exhaustive search over 14 substrates, every site orbit, $K = 2 \dots |V| + 2$, $r = 1 \dots K$ — **492 capacity transitions** — found: coker $\sigma_r = 0$ in all; $\dim D_r = \beta_1(Q_r)$ in all; reversible orbits self-loops in all; and **zero** cases of $c > 1$ with $X^{(r)}$ nonempty. The 48 cases with $c = 2$ all have $X^{(r)} = \emptyset$, which is the degenerate corner of Proposition 5.6.

This corrects a reading that circulated during the derivation: the saturated $c = 2$ example is not an “admissible but unreachable sector,” because there is nothing for it to be unreachable *from*. No non-degenerate disconnected layer is known.

5.3 The top layer

At $r = K - 1$ the layer is the single configuration m^K , so Q_{K-1} has two vertices and one edge per $\text{Stab}(m)$ -orbit on $N(m)$ — none reversible, since the endpoints are different configurations. Hence

$$\dim D_{K-1} = \#(N(m)/\text{Stab}(m)) - 1$$

zero exactly when the stabiliser is transitive on the neighbours of m . Verified 47/47.

This kills the reading that the ladder 11,4,0 terminates because capacity saturates. Every substrate in that ladder happened to have a locally transitive stabiliser. The prism at m of degree 3 with $|\text{Stab}| = 2$ gives $D_{K-1} = 1$; an interior path vertex with trivial stabiliser gives 1; the paw graph gives 2. **Local symmetry controls the terminal value; global saturation controls only the degenerate disconnection.** Different mechanisms.

6. Three levels

Forgetting the group gives the field-side defect $D_r^{\text{ord}} = H^1(X^{(r+1)}, X^{(r)})$; the observer-side defect is its invariant part. The difference

$$L = \dim D_r^{\text{ord}} - \dim D_r$$

is the sector that exists as transformation cycles but cannot be represented invariantly by this observer. For C_7 at $K = 4$: (25,9,1) ordinary against (11,4,0) invariant, so $L = (14,5,1)$. For Q_3 at $K = 4$: (69,20,2) against (11,3,0), $L = (58,17,2)$.

$$\text{admissibility } X^{(r)} \subset X^{(r+1)} \rightarrow \text{field-relative sector} \rightarrow \text{observer-visible sector}$$

Downward, σ_r looks like loss. Upward, the same object is newly exposed relative structure. The mathematics establishes the filtering; naming L “information loss” would require an independently defined observation map and is not done here.

7. Summary of status

statement	status
Lemma 1.1, Theorems 2.1–2.3	proved
$\epsilon = \beta_1(\mathcal{A}/G)$	proved (2.1 + 2.2)
$\mathcal{H}_G^1 = 0$	measured , 3 arenas $\times$ 2 predicates

statement	status
Theorem 3.1 reader bound	proved; attainment measured , 19 cases
extension / re-factorisation / hiding one-sided exactness	measured , 16 pairs; proofs sketched, not written
$\partial F = F \partial$ for subdivision	proved and verified to 0.0
Theorems 5.1–5.3, Lemmas 5.4–5.5, Prop. 5.6	proved
$\dim D_{K-1} = \#(N(m)/\text{Stab}(m)) - 1$	proved; 47/47
$c(Q_r) > 1$ only degenerately	proved (5.6); 492/492
contraction square $q_2^*: \ker \sigma \cong \ker \rho$	reported from a separate artifact; not reproduced by the code here
multigraph and filled-square-complex tables	excluded pending provenance

8. Corrections made during the derivation

These are recorded because each one is a distinction that the final statement depends on.

1. " ϵ measures creation of relational structure." False. Subdivision pullback is full rank on the cycle family: the channels pre-exist and the shorter cycle's group cannot see them.
2. " $\epsilon(X, 2) = 0$ universally." False — it held only for vertex-transitive members. $\epsilon(P_8, 2) = 6$, $\epsilon(P_6, 2) = 2$.
3. " $\$ = \$$ chiral-pair count." Matched at $K = 3$ for every N and fails from $K = 4$: $\epsilon(C_{12}, \cdot) = 9,32,63,77$ against chiral 9,14,28,30.
4. " $\text{Graded} \Leftrightarrow m \geq 3$ blocks and $\max \text{block} \geq K$." Fitted 28/28 at $K = 3$, broken at $K = 4$.
5. " $\text{The three-block threshold}$." Wrong side: the threshold is $K \geq 3$ tokens, a property of the system, not the reader.
6. " $\text{Grading requires a multi-token transition}$." False; single-token moves grade.
7. " $\text{Circulation on } Q_3 \text{ is evidence for the coexact reading}$." It was forced by $\epsilon = 1$; it confirmed the implementation, nothing more.
8. " $\text{As } G \rightarrow 1, \epsilon \rightarrow \beta_1$." There is no limit; the trivial group is an exact case.
9. " $\text{The capacity ladder terminates at } o \text{ because capacity saturates}$." False — Section 5.3.
10. " $\text{The saturated } c = 2 \text{ case is an admissible-but-unreachable sector}$." Degenerate: $X^{(r)}$ is empty there (Prop. 5.6).
11. A single-reader measurement that appeared to give a length-independent invariant $(\sum |Q|, \sum |a|) = (4,8)$ on both C_6 and C_8 did not survive sweeping all readers.
12. In preparing this paper, a test of the top-layer formula reported 16 failures out of 47. The failure was a destructuring bug in the test harness that forced the measured value to zero. The corrected test gives 47/47. Recorded because an uncorrected version of that table would have falsified a true theorem.

9. Verification

Self-contained. `itertools`, `numpy`, `collections` only.

The companion notebook `transformation_without_loss.ipynb` carries the complete implementation in a single self-contained library cell — `itertools`, `numpy` and `collections` only, no imports from other artifacts — followed by four verification cells. Running it top to bottom reproduces every number marked *measured* above. Its output is reproduced verbatim here.

$\epsilon(C8, K=3,4,5) = [2, 3, 2]$

$\epsilon(C_N, K=3) N=5..9 = [0, 0, 1, 2, 3]$

$\epsilon(K_N, \text{all } K) = [0, 0, 0, 0, 0]$

reader reach vs the bound $\epsilon(X,1)$:

$C8: \epsilon(X,1) = 0$

$C8 \quad K=3 \quad V=56 \quad E=120 \quad \epsilon=2 \quad \dim \pi^*C_1(X)=8 \quad \text{inv part}=0 \quad | \text{residual reached by readers: raw}=0 \text{ average } d=0$

$C8 \quad K=4 \quad V=70 \quad E=160 \quad \epsilon=3 \quad \dim \pi^*C_1(X)=8 \quad \text{inv part}=0 \quad | \text{residual reached by readers: raw}=0 \text{ average } d=0$

$Q3: \epsilon(X,1) = 0$

$Q3 \quad K=3 \quad V=56 \quad E=180 \quad \epsilon=1 \quad \dim \pi^*C_1(X)=12 \quad \text{inv part}=0 \quad | \text{residual reached by readers: raw}=0 \text{ averaged}=0$

$Petersen: \epsilon(X,1) = 0$

$Petersen \quad K=3 \quad V=120 \quad E=420 \quad \epsilon=1 \quad \dim \pi^*C_1(X)=15 \quad \text{inv part}=0 \quad | \text{residual reached by readers: raw}=0 \text{ averaged}=0$

$P8: \epsilon(X,1) = 0$

$P8 \quad K=3 \quad V=56 \quad E=105 \quad \epsilon=24 \quad \dim \pi^*C_1(X)=7 \quad \text{inv part}=3 \quad | \text{residual reached by readers: raw}=0 \text{ average } d=0$

$C7 \quad m=0 \quad K=4 \quad |Stab(m)|=2$

$r=1 \rightarrow 2: \dim R_{16 \rightarrow 27} \quad \text{coker}=0 \quad D_r=11 \quad | \quad Q_r: V=10 \quad E=20 \quad c=1 \quad \beta_1=11 \quad | \quad \text{ordinary}=25 \quad L=14 \quad \text{ok}$

$r=2 \rightarrow 3: \dim R_{27 \rightarrow 31} \quad \text{coker}=0 \quad D_r=4 \quad | \quad Q_r: V=4 \quad E=7 \quad c=1 \quad \beta_1=4 \quad | \quad \text{ordinary}=9 \quad L=5 \quad \text{ok}$

$r=3 \rightarrow 4: \dim R_{31 \rightarrow 31} \quad \text{coker}=0 \quad D_r=0 \quad | \quad Q_r: V=2 \quad E=1 \quad c=1 \quad \beta_1=0 \quad | \quad \text{ordinary}=1 \quad L=1 \quad \text{ok}$

$Q3 \quad m=0 \quad K=4 \quad |Stab(m)|=6$

$r=1 \rightarrow 2: \dim R_{25 \rightarrow 36} \quad \text{coker}=0 \quad D_r=11 \quad | \quad Q_r: V=7 \quad E=17 \quad c=1 \quad \beta_1=11 \quad | \quad \text{ordinary}=69 \quad L=58 \quad \text{ok}$

$r=2 \rightarrow 3: \dim R_{36 \rightarrow 39} \quad \text{coker}=0 \quad D_r=3 \quad | \quad Q_r: V=4 \quad E=6 \quad c=1 \quad \beta_1=3 \quad | \quad \text{ordinary}=20 \quad L=17 \quad \text{ok}$

$r=3 \rightarrow 4: \dim R_{39 \rightarrow 39} \quad \text{coker}=0 \quad D_r=0 \quad | \quad Q_r: V=2 \quad E=1 \quad c=1 \quad \beta_1=0 \quad | \quad \text{ordinary}=2 \quad L=2 \quad \text{ok}$

$prism \quad m=0 \quad K=4 \quad |Stab(m)|=2$

$r=1 \rightarrow 2: \dim R_{8 \rightarrow 20} \quad \text{coker}=0 \quad D_r=12 \quad | \quad Q_r: V=7 \quad E=18 \quad c=1 \quad \beta_1=12 \quad | \quad \text{ordinary}=26 \quad L=14 \quad \text{ok}$

$r=2 \rightarrow 3: \dim R_{20 \rightarrow 25} \quad \text{coker}=0 \quad D_r=5 \quad | \quad Q_r: V=4 \quad E=8 \quad c=1 \quad \beta_1=5 \quad | \quad \text{ordinary}=13 \quad L=8 \quad \text{ok}$

$r=3 \rightarrow 4: \dim R_{25 \rightarrow 26} \quad \text{coker}=0 \quad D_r=1 \quad | \quad Q_r: V=2 \quad E=2 \quad c=1 \quad \beta_1=1 \quad | \quad \text{ordinary}=2 \quad L=1 \quad \text{ok}$

$P6 \text{ (interior site)} \quad m=2 \quad K=4 \quad |Stab(m)|=1$

$r=1 \rightarrow 2: \dim R_{6 \rightarrow 17} \quad \text{coker}=0 \quad D_r=11 \quad | \quad Q_r: V=11 \quad E=21 \quad c=1 \quad \beta_1=11 \quad | \quad \text{ordinary}=11 \quad L=0 \quad \text{ok}$

$r=2 \rightarrow 3: \dim R_{17 \rightarrow 23} \quad \text{coker}=0 \quad D_r=6 \quad | \quad Q_r: V=6 \quad E=11 \quad c=1 \quad \beta_1=6 \quad | \quad \text{ordinary}=6 \quad L=0 \quad \text{ok}$

$r=3 \rightarrow 4: \dim R_{23 \rightarrow 24} \quad \text{coker}=0 \quad D_r=1 \quad | \quad Q_r: V=2 \quad E=2 \quad c=1 \quad \beta_1=1 \quad | \quad \text{ordinary}=1 \quad L=0 \quad \text{ok}$

top layer: $D_{(K-1)} = \#(N(m)/Stab(m)) - 1$

nonzero: $P_4 \quad m=1 \quad K=3 \quad \deg=2 \quad |Stab|=1 \quad \text{orbits}=2 \rightarrow D=1$

nonzero: $P_4 \quad m=1 \quad K=4 \quad \deg=2 \quad |Stab|=1 \quad \text{orbits}=2 \rightarrow D=1$

nonzero: P5 m=1 K=3 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: P5 m=1 K=4 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: P6 m=1 K=3 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: P6 m=1 K=4 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: P6 m=2 K=3 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: P6 m=2 K=4 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: prism m=0 K=3 deg=3 |Stab|=2 orbits=2 -> D=1
nonzero: prism m=0 K=4 deg=3 |Stab|=2 orbits=2 -> D=1
nonzero: tri+tail m=0 K=3 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: tri+tail m=0 K=4 deg=2 |Stab|=1 orbits=2 -> D=1
nonzero: tri+tail m=2 K=3 deg=3 |Stab|=2 orbits=2 -> D=1
nonzero: tri+tail m=2 K=4 deg=3 |Stab|=2 orbits=2 -> D=1
nonzero: paw m=0 K=3 deg=3 |Stab|=1 orbits=3 -> D=2
nonzero: paw m=0 K=4 deg=3 |Stab|=1 orbits=3 -> D=2
 47/47 agree

saturation $K=|V|+r$ forces the old arena empty:

$C_3 \quad |V|=3 \ r=1 \ K=4: |X^\wedge(r)| = 0$
 $C_3 \quad |V|=3 \ r=2 \ K=5: |X^\wedge(r)| = 0$
 $C_4 \quad |V|=4 \ r=1 \ K=5: |X^\wedge(r)| = 0$
 $C_4 \quad |V|=4 \ r=2 \ K=6: |X^\wedge(r)| = 0$
 $C_5 \quad |V|=5 \ r=1 \ K=6: |X^\wedge(r)| = 0$
 $C_5 \quad |V|=5 \ r=2 \ K=7: |X^\wedge(r)| = 0$
 $C_6 \quad |V|=6 \ r=1 \ K=7: |X^\wedge(r)| = 0$
 $C_6 \quad |V|=6 \ r=2 \ K=8: |X^\wedge(r)| = 0$
 $C_7 \quad |V|=7 \ r=1 \ K=8: |X^\wedge(r)| = 0$
 $C_7 \quad |V|=7 \ r=2 \ K=9: |X^\wedge(r)| = 0$

Reading the capacity block: $\dim R$ is the invariant residual dimension either side of the transition, **coker** is the cokernel of σ_r (Theorem 5.1 predicts 0), D_r is the capacity defect measured by linear algebra, the Q_r columns are the cone quotient graph, **beta1** is Theorem 5.3's prediction, **ordinary** is the same defect with the group forgotten, and L is the difference of Section 6. The prism row is the counterexample to terminal exhaustion: $D_3 = 1$. The interior-site path row has trivial stabiliser, so $L = 0$ throughout — field and observer agree exactly when there is no symmetry to filter with.

10. Boundary of the result

These are not disclaimers. Each records a distinction established during the derivation, and each prevents two mathematically different objects from being identified because they behaved similarly on some slice of the parameter space. The boundary is part of the content.

$D_r \neq$ **reader information loss**. D_r is a relative invariant cohomology group. Calling the difference L of Section 6 “loss” requires an independently defined observation map; Theorem 3.1 shows the readers actually constructed here see $\epsilon(X, 1)$ of it and no more.

$D_r \neq$ **mass, energy, or price**. No dimensional map has been constructed. p_r is an integer sequence of Betti numbers.

$D_{K-1} = 0 \neq$ **price exhaustion**. Section 5.3 gives the exact cause: local stabiliser transitivity at m .

New states $\neq$ new channels. For C_7 at $K = 4$ the layers add 15,6,1 configurations, carry 25,9,1 ordinary relative cycles, and contribute 11,4,0 invariant channels. Three different quantities; only the last is p_r .

Symmetry filtering $\neq$ destruction. $R_{G'} \hookrightarrow R_G$ is injective. What a larger group cannot represent invariantly is still present.

Change $\neq$ loss. Loss = change + removal of admissible continuation, and even then the removed sector is a relative class, not an annihilation.

ϵ **is not a base-graph invariant**. It is not controlled by degree, bipartiteness, $|V|$, $|E|$, or girth. C_6 (bipartite) gives 0, C_8 (bipartite) gives 2, Petersen (non-bipartite) gives 1. It measures how much loop structure survives the quotient — a property of the group’s grip on the transformation set.

There is still no summary formula for $\epsilon(C_N, K)$. The closed form of Theorem 2.3 computes it; nothing compresses it.

11. Interpretation

This section proves nothing and is separable from the rest of the paper.

The results have a consistent shape. Every elementary change to the substrate or the observer either adds transformation structure, re-expresses it, or removes it from view. The only operation that produces a kernel is a restriction of which states are admissible, and the kernel it produces is a relative class: the transformations are present in the larger admissible space and absent from the smaller one. In this setting, therefore:

change does not entail loss; loss = change + removal of admissible continuation.

If one is looking for a notion of cost attached to transformation, the candidate this work produces is not a scalar. It is the graded sequence $p_r = \beta_1(Q_r)$ indexed by how much admissibility has been relaxed, and its value is not the number of states admitted but the number of independent transformation cycles they carry after exact and symmetry-trivial structure has been quotiented away. Whether that has any physical referent is not addressed here.

The programme this belongs to holds that distinction requires lawful continuation and that no state is terminal. The mathematics above is consistent with that and does not establish it. What it does establish is narrower and, for that reason, usable: in a fully specified finite model of tokens moving on a graph, the structure of transformation is conserved under every change of the rules of movement except a restriction of what may exist, and that exception is measured exactly.

12. Open

1. Prove the one-sided exactness statements of Section 4. All three look short; none is written.
 2. Find $\mathcal{H}_G^1 \neq 0$, or prove it cannot occur. This is the single result that would reopen a non-coexact channel. Candidates: high-girth substrates with small automorphism groups.
 3. A recurrence for p_r . Setting $k = K - r - 1$, the layer's internal part is the ordinary k -token arena on $X - \{m\}$ with a cone attached wherever $N(m) \not\subseteq U$, so $p_r = P_X(K - r - 1)$ for a single function P_X . Whether P_X satisfies a recurrence in k is the natural next target.
 4. A non-degenerate disconnected layer, under a modified admissibility rule (disconnected substrate, forbidden moves, or capacity below one elsewhere). Proposition 5.6 shows the present rules cannot produce one.
 5. Reproduce or retract the multigraph and filled-square-complex tables excluded in Section 7.
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Reproducibility

Every number in Sections 2–6 marked *measured* is produced by the code in Section 9 or by the companion notebooks in the same repository, with the single exception noted in Section 7. All ranks use an absolute singular-value tolerance of 10^{-9} ; a relative tolerance reports full rank on a matrix whose entries are 10^{-16} , and that error was made once during this work.

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